

Total stock: cycle, safety, process, pipeline and anticipation stock

Part 2 · Categories of stock — solution report with data, method, results and charts

Contents

1. Part 2 · Categories of stock — Total stock: cycle, safety, process, pipeline and anticipation stock

1. Total stock: cycle, safety, process, pipeline and anticipation stock

Pipeline stock = daily demand × transit days (Little's law); Cycle = $\frac{1}{2} \times$ cycle days

\$331,095

Total inventory value

23.1 %

Safety share of value

P4 Softener (36.0 days)

Highest days of cover

18.5 %

Pipeline share of value

33.4 %

Cycle share of value

10.5 %

Process share of value

14.5 %

Anticipation share of value

Total stock broken into its functional categories

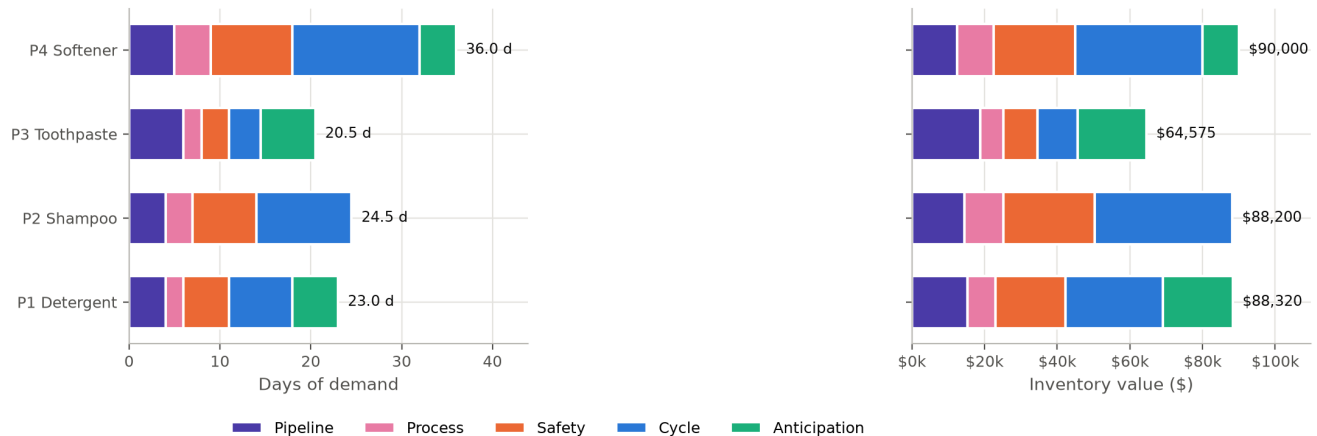


Figure 1.1 — Stacked stock components per product, in days of demand (left) and in value (right).

Stock profile over time — P1 Detergent

Pipeline, process and safety stock form a fixed base; cycle stock saw-tooths on top; anticipation stock is built ahead of a peak

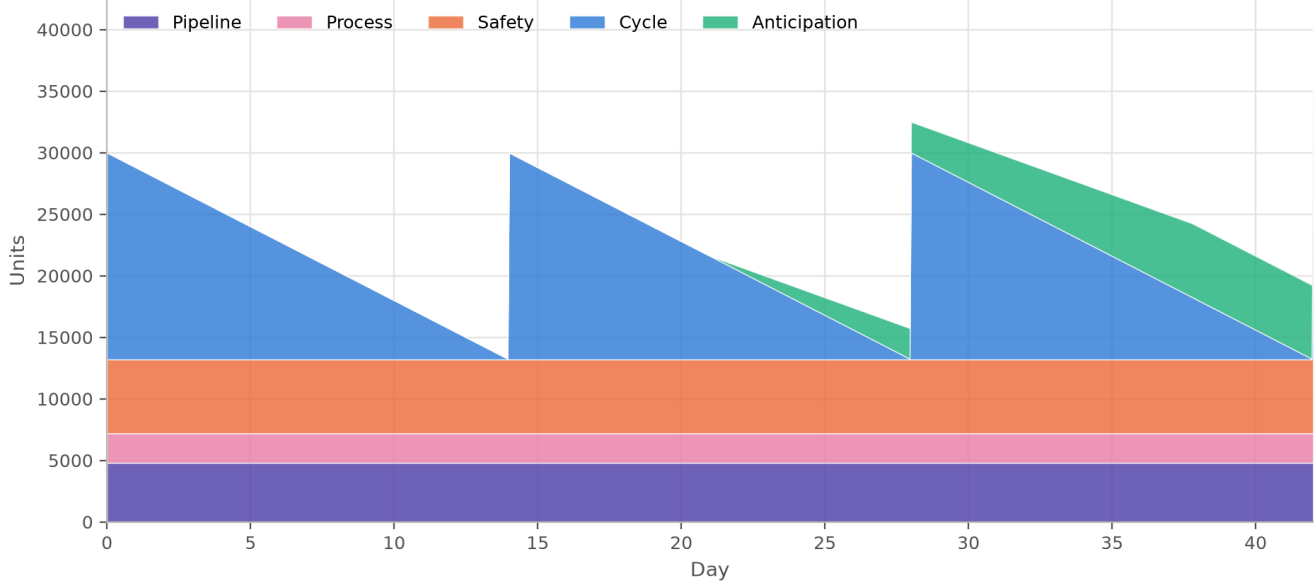


Figure 1.2 — Layered inventory profile (illustrative anticipation build in the last third of the horizon).

Cycle-stock coverage is directly proportional to the production cycle

Slope $\frac{1}{2}$: every extra day between runs adds half a day of stock; the intercept is the fixed stock

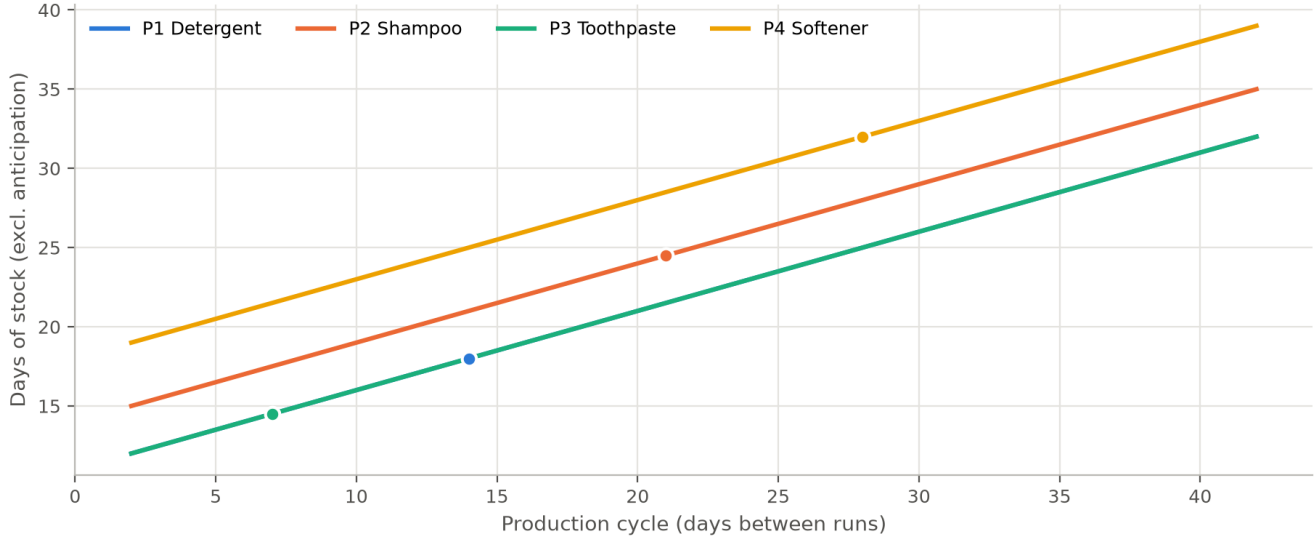


Figure 1.3 — Sensitivity of total days of stock to the production cycle length. Dots = current cycles.

Stock components in days of demand.

product	Pipeline	Process	Safety	Cycle	Anticipation	Total
P1 Detergent	4.0	2.0	5.0	7.0	5.0	23.0
P2 Shampoo	4.0	3.0	7.0	10.5	0.0	24.5
P3 Toothpaste	6.0	2.0	3.0	3.5	6.0	20.5
P4 Softener	5.0	4.0	9.0	14.0	4.0	36.0

Stock components in units.

product	Pipeline	Process	Safety	Cycle	Anticipation	Total
P1 Detergent	4,800	2,400	6,000	8,400	6,000	27,600
P2 Shampoo	3,200	2,400	5,600	8,400	0	19,600
P3 Toothpaste	9,000	3,000	4,500	5,250	9,000	30,750
P4 Softener	2,500	2,000	4,500	7,000	2,000	18,000

Stock components in value (\$).

product	Pipeline	Process	Safety	Cycle	Anticipation	Total
P1 Detergent	15,360	7,680	19,200	26,880	19,200	88,320
P2 Shampoo	14,400	10,800	25,200	37,800	0	88,200
P3 Toothpaste	18,900	6,300	9,450	11,025	18,900	64,575
P4 Softener	12,500	10,000	22,500	35,000	10,000	90,000